

Homework 3 Solutions

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Problem 8.1

AA, AE, AI, AO, AU, EA, EE, EI, EO, EU, IA, IE, II, IO, IU, OA, OE, OI, OO, OU, UA, UE, UI, UO, UU.

Problem 8.3

For a bit string of length 1, you have 2 strings "0" or "1". For a bit string of length 2, you have 4 strings "00", "01", "10", or "11". For a bit string of length 3, you have 8 strings "000", "001", "010", "011", "100", "101", "110", or "111". Notice that for each position in the bit string, you have 2 options (either a 1 or a 0), and that a length- k bit string has k positions to fill. Thus the number of possible strings is $2 \times 2 \times 2 \times \dots \times 2$ a total of k times, or in other words, you have 2^k possible strings.

Problem 8.7

a) $31!$

Notice that there are a total of $30!$ ways to arrange the photos in order, and then once a particular arrangement has been chosen, there are 31 ways I could divide the photos into the two pages. For example if I only had 2 photos, call them A and B , then there would be $2!$ ways to arrange them (AB or BA), and then once I chose an arrangement (say AB) then there would be 3 ways to divide the photos into the two pages (all on the first page AB), one on each page $A|B$, or both on the second page $|AB$). Thus, for 30 photos, there's $30! \times 31$ or $31!$ possible arrangements keeping track of order.

b) 2^{30}

Notice that for each of the 30 photos I have two options, either the photo goes on the first page or the photo goes on the second page. Since I have to make a decision with 2 possible outcomes 30 times, then there are 2^{30} possible outcomes. Notice that we could also interpret this problem as "How many binary string scan I make with a length of 30?". Thinking about it this way, a "0" in the first position would mean the first photo goes on the first page, and a "1" in the first position would mean the first photo goes on the second page. We could then repeat the process for the remaining 29 positions for the other 29 photos. Thus, once again, the total number is 2^{30} .

Problem 8.12

a) 10^9

b) 5×10^8

c) 5^9

d) 10^5

e) 9^9

f) $10^9 - 9^9$

g) 9^9

Problem 8.13

a) *Proof.* $n^2 = (n)_2 + n$ algebraically.

1. Observe that $(n)_2 = n(n-1)$.
2. Then $(n)_2 + n = n(n-1) + n = n^2 - n + n = n^2$.
3. Therefore $n^2 = (n)_2 + n$.

□

b) *Proof.* $n^2 = (n)_2 + n$ by counting.

1. Consider all the ordered pairs (x, y) we could construct from the set of numbers $\{1, 2, 3, \dots, n\}$.
2. Since we would have n options for x element and n options for the y element, then we have a total of n^2 possible outcomes.
3. Thus n^2 is the count of all possible ordered pairs, and it remains to show that $(n)_2 + n$ is also the count.
4. Observe that in counting all the ordered pairs, we could first count all the pairs without any repeating elements and then count the pairs with repeating elements, thereby counting all the pairs.
5. Observe that when creating an ordered pair without any repetitions then we have n options for the first element, but only $n-1$ options for the second (since we cannot choose the same number we used for the first element).
6. Thus $n(n-1) = (n)_2$ is the count of ordered pairs without repeats.
7. Observe that when creating an ordered pair with only repetitions then we have n options for the first element, but only 1 options for the second (since we have to choose the same number we used for the first element).
8. Thus $n \cdot 1 = n$ is the count of ordered pairs with repeats.
9. Thus $(n)_2 + n$ is the count of all the ordered pairs.
10. Therefore, since both n^2 and $(n)_2 + n$ count the same objects, then $n^2 = (n)_2 + n$.

□

Problem 8.19

$52 \cdot 36 \cdot 22 \cdot 10$